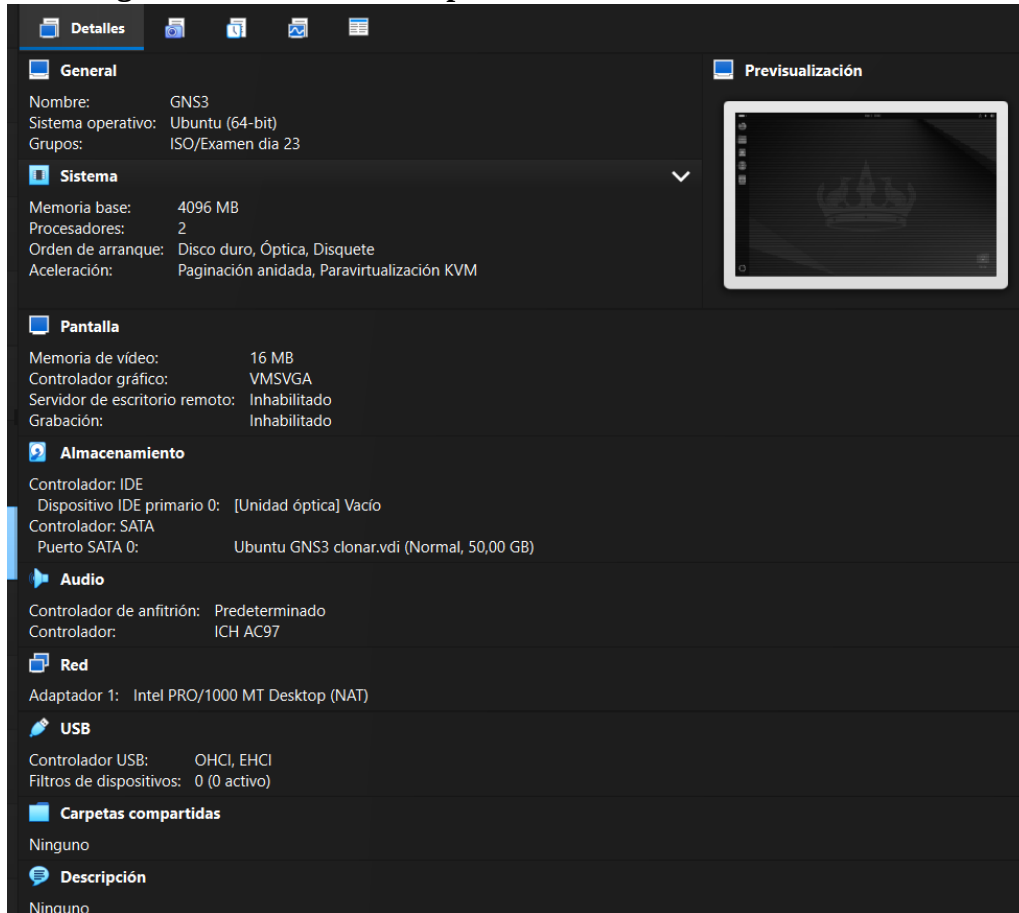
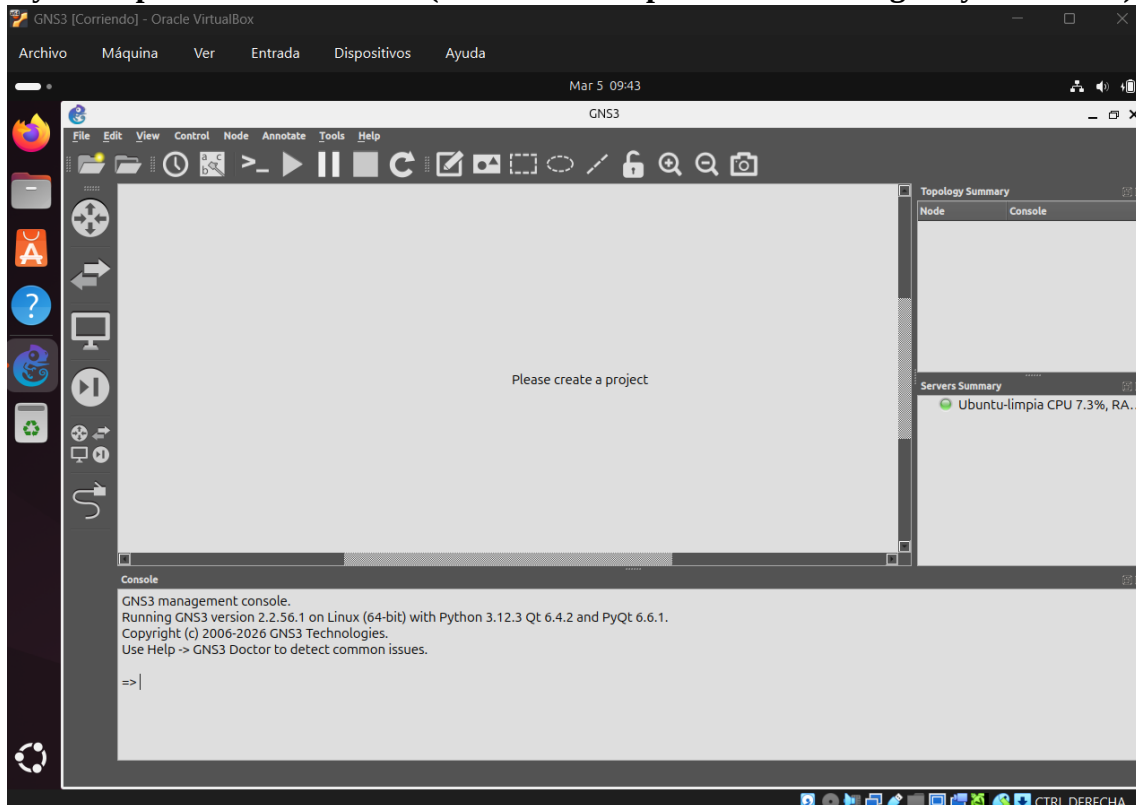


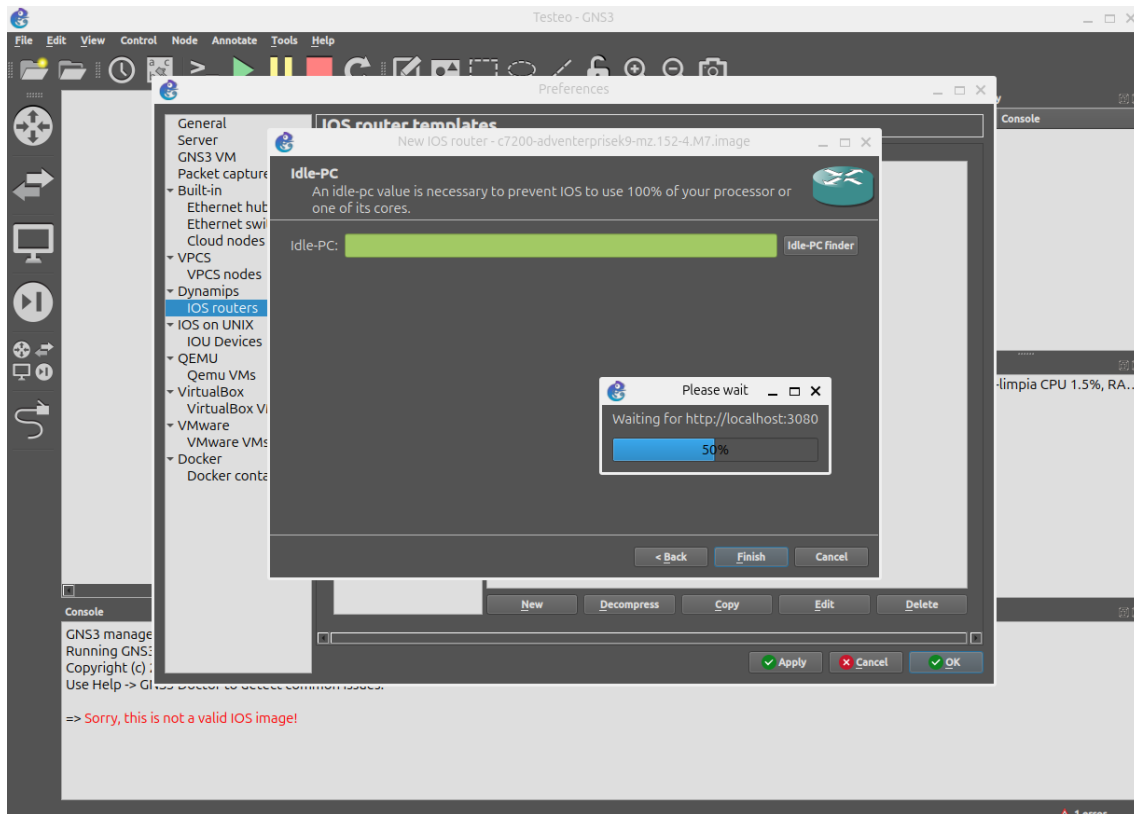
1. Configuración de VirtualBox para GNS3



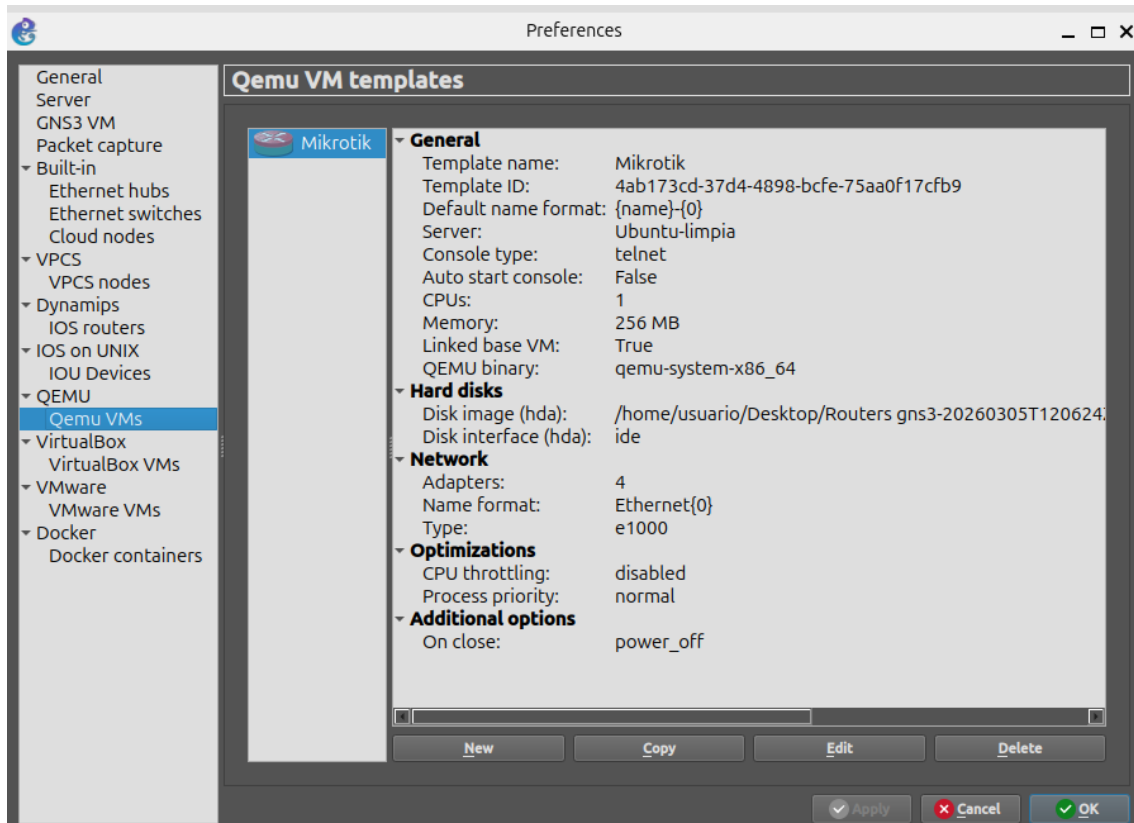
2 y 3. Preparación del Sistema (Ubuntu Desktop e instalación de gns3 y wireshark)



Añadimos el router Cisco en GNS3 metiendo la ISO



Mikrotik

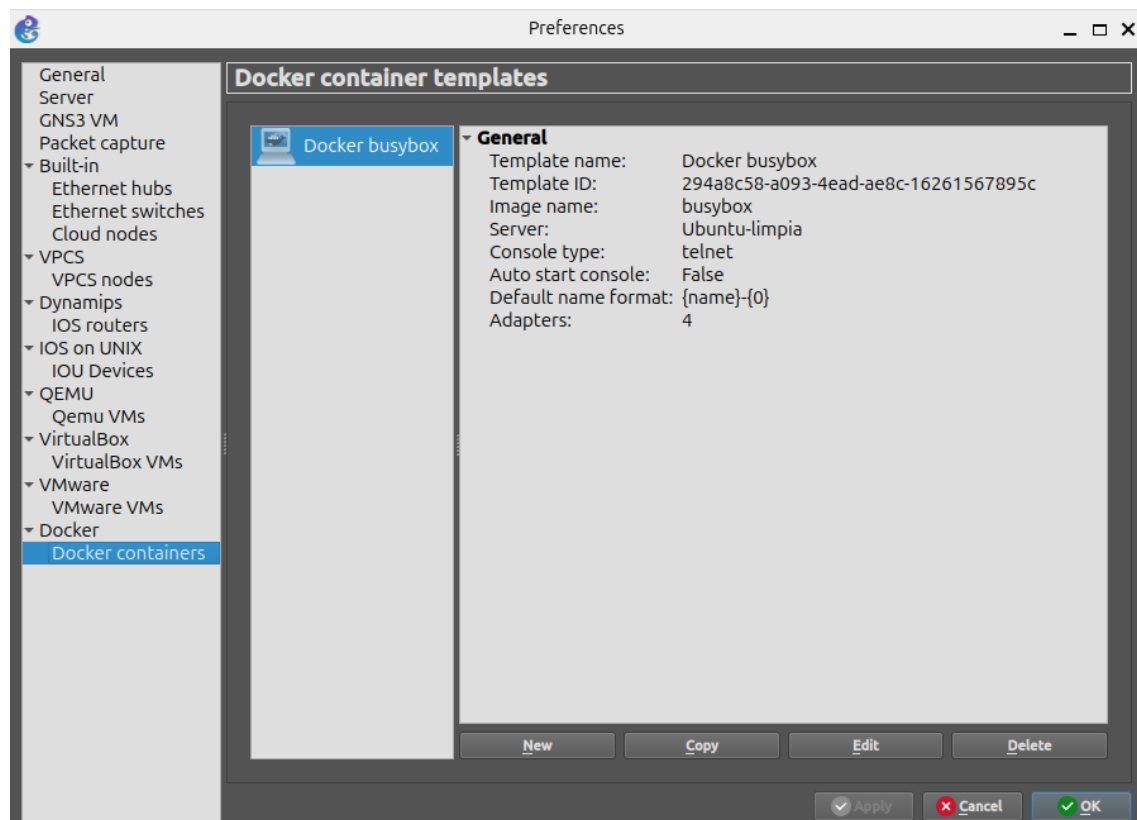


Instalacion de dockers

```
usuario@Ubuntu-limpia:~$ docker version
Client: Docker Engine - Community
 Version: 29.2.1
 API version: 1.53
 Go version: go1.25.6
 Git commit: a5c7197
 Built: Mon Feb  2 17:17:26 2026
 OS/Arch: linux/amd64
 Context: default

Server: Docker Engine - Community
 Engine:
  Version: 29.2.1
  API version: 1.53 (minimum version 1.44)
  Go version: go1.25.6
  Git commit: 6bc6209
  Built: Mon Feb  2 17:17:26 2026
  OS/Arch: linux/amd64
  Experimental: false
 containerd:
  Version: v2.2.1
  GitCommit: dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version: 1.3.4
  GitCommit: v1.3.4-0-gd6d73eb8
 docker-init:
  Version: 0.19.0
  GitCommit: de40ad0
usuario@Ubuntu-limpia:~$
```

Instalacion de docker busybox en gns3



Practica de enrutamiento estático en GNS3

Gestion del router

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
```

```
R1(config)#interface f1/0
R1(config-if)#ip address 192.168.20.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  9 12:04:32.723: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar  9 12:04:33.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R1(config)#do wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]S
No action taken because command was not confirmed
R1(config)#
```

Docker

```
Dockerbusybox-2
Trying ::1...
Connected to localhost.
Escape character is '^['.

/ # ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: seq=0 ttl=255 time=33.744 ms
64 bytes from 192.168.10.1: seq=1 ttl=255 time=8.492 ms
64 bytes from 192.168.10.1: seq=2 ttl=255 time=10.599 ms
^C
--- 192.168.10.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 8.492/17.611/33.744 ms
/ # ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1): 56 data bytes
64 bytes from 192.168.20.1: seq=0 ttl=255 time=10.980 ms
64 bytes from 192.168.20.1: seq=1 ttl=255 time=2.909 ms
^C
--- 192.168.20.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.909/6.944/10.980 ms
/ #
```

NAT/PAT

```
R1
R1(config-if)#exit
R1(config)#ip nat inside
*Mar  9 12:25:08.343: %DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet1/1 assigned
DHCP address 10.0.2.16, mask 255.255.255.0, hostname R1

R1(config)#int f1/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#do wr
Building configuration...
[OK]
R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#access-list 1 permit 192.168.20.0 0.0.0.255
R1(config)# ip nat inside source list 1 interface f1/0 overload
R1(config)#ping 8.8.8.8
      ^
% Invalid input detected at '^' marker.

R1(config)#do ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/24/28 ms
R1(config)#
```

NGINX

Los repositorios de alpine se quedan colgados constantemente

```
alpine-1
/ # apk update
/ # apk add nginx
/ # mkdir -p /run/nginx
/ # nginx
/bin/sh: nginx: not found
/ # apk update --no-cache
/ # ip link set dev eth0 mtu 1400
/ # echo "http://dl-cdn.alpinelinux.org/alpine/v3.18/main" > /etc/apk/repositories
/bin/sh: can't create /etc/apk/repositories: nonexistent directory
/ # echo "http://dl-cdn.alpinelinux.org/alpine/v3.18/community" >> /etc/apk/repositories
/bin/sh: can't create /etc/apk/repositories: nonexistent directory
/ # apk update
[fetch https://dl-cdn.alpinelinux.org/alpine/v3.23/main/x86_64/APKINDEX.ta...]
/ # apk update
[fetch https://dl-cdn.alpinelinux.org/alpine/v3.23/main/x86_64/APKINDEX.ta...]S^
[[H
/ # ip link set dev eth0 mtu 1400
/ # sed -i 's/https/http/g' /etc/apk/repositories
/ # apk update
/ # date -s "2026-03-09 13:45:00"
Mon Mar  9 13:45:00 UTC 2026
/ # apk update
/ #
```

Pulling 'busybox:latest' from Docker repository

Análisis con Wireshark

